
Towards auction-based multi-agent collision-avoidance under continuous stochastic dynamics

Jan-P. Calliess, Michael Osborne & Stephen J. Roberts

JAN, MOSB, SJROB@ROBOTS.OX.AC.UK

Department of Engineering Science, University of Oxford, Parks Road, Oxford, OX1 3PJ, UK

1. Overview

We describe an approach to multi-agent planning under continuous stochastic dynamics. The approach yields collision-free state trajectories with adjustably high certainty, while aiming for low social cost. To this end we describe a collision-detection module based on a distribution-independent probabilistic bound. We employ an optimization-based approach to incrementally alter plans until all collisions are avoided with sufficiently high confidence. We consider the case of feedback controlled agents, and alter plans by introducing new setpoints to the agents' controllers. In the context of a simple stochastic path planning scenario, we compare two alternative strategies for finding such setpoints.

Due to their practical importance, multi-agent collision avoidance and control have been extensively studied across different communities including AI, robotics and control. However, these works typically assume linear and discrete dynamic models; by contrast, our work intends to overcome these limitations and to present solutions for continuous state space. While our current experiments were conducted with linear stochastic differential equation (SDE) models with state-independent noise (yielding Gaussian processes) our method is also applicable to non-Gaussian cases with state-dependent uncertainties.

To ensure collision avoidance yields low social cost across the entire agent collective, we compare two different coordination mechanisms. Firstly, we consider a simple fixed-priority scheme Erdmann & Lozano-Perez (1987), and secondly, we modify an auction-based coordination protocol Calliess et al. (2011) to work in our continuous setting. In contrast to pre-existing work in auction-style multi-agent planning (e.g. Tovey et al. (2005); Lagoudakis et al. (2005); Calliess et al. (2011)) and multi-agent collision avoidance (e.g. Kostic et al. (2010); Ayanian & Kumar (2010)), we avoid *a priori* discretizations of space and time and present solutions for continuous time and state space.

2. Method

In this section we outline the core ideas of our proposed approach to multi-agent collision avoidance. After specifying the agents' dynamics and formalizing the notion of a single-agent plan, we define the multi-agent planning task. In Sec. 2.2, we describe the two coordination approaches we consider utilizing to generate conflict-free plans. During coordination, collision detection and conflict resolution are key. Sec. 2.1 contains an exposition of our proposed methods to achieve both.

Model. We assume the system contains a set $\mathfrak{A}$ of agents indexed by $\mathbf{a} \in \{1, \dots, |\mathfrak{A}|\}$. Each agent $\mathbf{a}$'s associated plant has a probabilistic state trajectory following a stochastic controlled D -dimensional state dynamics (we consider the case $D = 2$) in the continuous interval of (future) time $I = (t_0, t_f]$. For our method to work all we need to require is that the trajectory's mean function $m : I \rightarrow \mathbb{R}^D$ and covariance matrix function $\Sigma : I \rightarrow \mathbb{R}^{D \times D}$ are evaluable for all times $t \in I$. A prominent class for which closed-form moments can be easily derived are linear Ito-SDEs. For instance, we consider the SDE

$$dx^{\mathbf{a}}(t) = K(\xi^{\mathbf{a}}(t) - x^{\mathbf{a}}(t))dt + B dW \quad (1)$$

where $K, B \in \mathbb{R}^{D \times D}$ are matrices $x^{\mathbf{a}} : I \rightarrow \mathbb{R}^D$ is the state trajectory and W is a vector-valued Wiener process. Here, $u(x^{\mathbf{a}}) := K(\xi^{\mathbf{a}} - x^{\mathbf{a}})$ will be interpreted as the control output of a linear feedback-controller regulating the state to converge to a desired signal $\xi^{\mathbf{a}}(t) = \zeta_0^{\mathbf{a}} \chi_{\{0\}}(t) + \sum_{i=1}^{H^{\mathbf{a}}} \zeta_i^{\mathbf{a}} \chi_{\tau_i^{\mathbf{a}}}(t)$ where $\chi_{\tau_i} : \mathbb{R} \rightarrow \{0, 1\}$ denotes the indicator function of the half-open interval $\tau_i^{\mathbf{a}} = (t_{i-1}^{\mathbf{a}}, t_i^{\mathbf{a}}] \subset [0, T^{\mathbf{a}}]$ and each $\zeta_i^{\mathbf{a}} \in \mathbb{R}^D$ is a *setpoint*. If K is positive definite the agent's state trajectory is determined by setpoint sequence $p^{\mathbf{a}} = (t_i^{\mathbf{a}}, \zeta_i^{\mathbf{a}})_{i=0}^{H^{\mathbf{a}}}$ (aside from the random disturbances) which we will refer to as the agent's *plan*. For example, plan $p^{\mathbf{a}} := ((t_0, x_0^{\mathbf{a}}), (t_f, x_f^{\mathbf{a}}))$ could be used to regulate agent $\mathbf{a}$'s *start state* $x_0^{\mathbf{a}}$ to a given *goal state* $x_f^{\mathbf{a}}$ between times t_0 and t_f . For simplicity, we assume the agents are always initialized with plans of

this form before coordination commences.

Roughly speaking, we can interpret the controlled agent's behavior as per Eq. 1 as follows: x^a describes the location of the a th agent, which is trying to follow a prescribed course given by ξ^a . At each time t_i^a , the agent heads towards a new location ζ_{i+1}^a (the setpoint). Since the agent's position is subject to other random disturbances (corresponding to the noise $B dW$), the agent needs to correct the course proportionally to the deviation from the prescribed course. This correcting activity corresponds to the control term $K(\xi^a(t) - x^a(t))dt$.

Task. Each agent a desires to find a sequence of setpoints (p^a) such that (i) it moves from its start state x_0^a to its goal state x_f^a along a low-cost trajectory and (ii) such that along the trajectory its plant (with diameter Δ) does not collide with any other agents' plant in state space with at least a given probability $1 - \delta \in (0, 1)$.

2.1. Collision detection and resolution

To ensure setpoint-plans yield collision-free trajectories, the two main mechanisms to develop are (I) an algorithm capable of detecting collisions on a *continuous* time interval I with a sufficiently high probability and (II) to use this information to augment the existing plan by new setpoints in order to avoid the collision with the given confidence bound $1 - \delta$.

Next, we will briefly sketch the underlying ideas of our approach to address the two problems.

(I) **Collision detection.** We employ a bound for the probability of a collision at any point to conservatively identify probable collisions. Based on a generalized Chebychev-bound (Whittle, 1958), the authors of Lyons et al. (2012) derived D distribution-independent constraints of the form $C_j(t) > 0$ ($j = 1, \dots, D$). These depend on the components of the means and the covariance matrix functions of two stochastic trajectories, as well as on δ . For any given time t , it was shown that satisfaction of at least one of the constraints is a sufficient condition for collision avoidance (with prob. $> 1 - \delta$). For conservative collision detection between two agents' trajectories x^a, x^r , we propose to construct a (continuous) *criterion function* $\gamma^{a,r}(t) := \max_j \{C_j(t)\}$. By construction, a collision between the trajectories with probability above δ can be ruled-out if $\gamma^{a,r}$ attains only positive values. To determine the presence or absence of a collision with any other agent, agent a can minimize the continuous function $\Gamma^a(t) := \min_{r \neq a} \{\gamma^{a,r}(t)\}$ employing a suitable numerical optimizer. Being conservative we as-

sume a collision prevails unless $\min_{t \in I} \Gamma^a(t) > 0$.

(II) **Collision resolution.** An agent seeks to avoid collisions by adding new setpoints to its plan until the maximum instantaneous collision probability of the resulting state trajectory drops below threshold δ . For choosing these new setpoints we consider two methods **WAIT** and **FREE**. In the first method, the agent inserts a time-setpoint pair (t, x_0^a) into the previous plan p^a . Since this aims to cause the agent to wait at its start location x_0^a we will call the method **WAIT**. It is possible that multiple such insertions are necessary until collisions are avoided. Of course, if a higher-priority agent decides to pass through x_0^a , this method is too rigid to resolve a conflict. Therefore, we propose a second method. Therein, the agent optimizes for the time and location of the new setpoint. Let $p_{\uparrow(t,s)}^a$ be the plan updated by insertion of time-setpoint pair $(t, s) \in I \times \mathbb{R}^D$. We propose to choose the candidate setpoint (t, s) that minimizes a function being a weighted sum of the expected cost entailed by executing updated plan $p_{\uparrow(t,s)}^a$ and a collision penalty $-\lambda \min_t \Gamma^a(t) H(-\min_t \Gamma^a(t))$. Here, λ is a large number, H is the Heaviside function and Γ^a is computed based on the assumption we were to execute $p_{\uparrow(t,s)}^a$. Since the new setpoint can be chosen freely in time and state-space we refer to the method as **FREE**.

2.2. Coordination

We will now consider how to integrate our collision detection and avoidance methods into a coordination framework that determines who needs to avoid whom and at what stage of the coordination process. Such decisions are known to significantly impact the *social cost* (i.e. the sum of all agents' individual costs) of the agent collective.

Fixed-priorities (FP). As a baseline method for coordination we consider a basic fixed-priority method (e.g. Erdmann & Lozano-Perez (1987); Bennewitz et al. (2001)). Here, each agent has a unique ranking (or priority) according to its index a (i.e. agent 1 has highest priority, agent $|A|$ lowest). When all higher-ranking agents are done planning, agent a is informed of their planned trajectories which it has to avoid with a probability greater than $1 - \delta$.

Lazy Auction Protocol (AUC). While the FP method is simple and fast the rigidity of the fixed ranking can lead to sub-optimality in terms of social cost. To alleviate this we propose to revert the ranking flexibly on a case-by-case basis. In particular, the agents are allowed to compete for the right to gain passage (e.g. across a region where a collision was detected)

input : Agents $\mathbf{a} \in \mathcal{A}$, cost functions $c^{\mathbf{a}}$, dynamics, initial start and goal states, initial plans $p^1, \dots, p^{|\mathcal{A}|}$.
output: collision-free plans $p^1, \dots, p^{|\mathcal{A}|}$.
repeat
 for $\mathbf{a} \in \mathcal{A}$ **do**
 $[\text{flag}^{\mathbf{a}}, \mathcal{C}^{\mathbf{a}}, t_{\text{coll}}] \leftarrow \text{CollDetect}^{\mathbf{a}}(\mathbf{a}, \mathcal{A} - \{\mathbf{a}\})$
 if $\text{flag}^{\mathbf{a}} = 1$ **then**
 $\text{winner} \leftarrow \text{Auction}(\mathcal{C}^{\mathbf{a}} \cup \{\mathbf{a}\}, t_{\text{coll}})$
 foreach $\mathbf{r} \in (\mathcal{C}^{\mathbf{a}} \cup \{\mathbf{a}\}) - \{\text{winner}\}$ **do**
 $p^{\mathbf{r}} \leftarrow \text{Avoid}^{\mathbf{r}}((\mathcal{C}^{\mathbf{a}} \cup \{\mathbf{a}\}) - \{\mathbf{r}\}, t_{\text{coll}})$
 $\text{Broadcast}^{\mathbf{r}}(p^{\mathbf{r}})$
 end
 end
 end
until $\forall \mathbf{a} \in \mathcal{A} : \text{flag}^{\mathbf{a}} = 0$;

Figure 1. Lazy auction coordination method (AUC) (written in a sequentialized form). Collisions are resolved by choosing new setpoints to enforce collision avoidance. $\mathcal{C}^{\mathbf{a}}$: set of agents detected to be in conflict with $\mathbf{a}$. $\text{flag}^{\mathbf{a}}$: collision detection flag ($=0$, iff no collision detected). t_{coll} : earliest time where a collision was detected. **Avoid**: collision resolution method updating the plan by a single new setpoint according to WAIT or FREE (ref. (II), Sec. 2.1).

by submitting bids in the course of an auction. The structure of the approach is outlined in Fig. 1.

Assume an agent $\mathbf{a}$ detects a collision at a particular time step t_{coll} and invites the set of agents $\mathcal{C}^{\mathbf{a}} = \{\mathbf{r} | \gamma^{\mathbf{a}, \mathbf{r}}(t_{\text{coll}}) \leq 0\}$ to join an auction to decide who needs to avoid whom. In particular, the auction determines a winner who is not required to alter its plan. The losing agents need to insert a new setpoint into their respective plans designed to avoid all other agents in $\mathcal{C}^{\mathbf{a}}$ while keeping the plan cost function low.

The idea is to design the auction rules as a heuristic method to minimize social cost of the ensuing solution. To this end, we define the bids to be approximately proportional to the expected regret for losing and not gaining passage. That is, agent $\mathbf{a}$ submits a bid $b^{\mathbf{a}} = l^{\mathbf{a}} - s^{\mathbf{a}}$. $l^{\mathbf{a}}$ specifies $\mathbf{a}$'s anticipated cost for the event that the agent will not secure “the right of passage” and has to create a new setpoint (according to (II) in Sec. 2.1), tailored to avoid all other agents engaged in the current auction. On the other hand, $s^{\mathbf{a}}$ is the cost of the unchanged plan $p^{\mathbf{a}}$. If there is a tie among multiple agents the agent with the lowest index among the highest bidders wins.

Acknowledging that $s^{\text{winner}} + \sum_{\mathbf{a} \in \text{losers}} l^{\mathbf{a}}$ is an estimated social cost (based on current beliefs of trajectories) after the auction, we see that the winner determination rule greedily attempts to minimize social cost: $b^{\text{winner}} \geq b^{\mathbf{r}} \Leftrightarrow \forall \mathbf{r} : s^{\mathbf{r}} + \sum_{\mathbf{a} \neq \mathbf{r}} l^{\mathbf{a}} \geq s^{\text{winner}} + \sum_{\mathbf{a} \neq \text{winner}} l^{\mathbf{a}}$.

3. Simulations

As a first test, we simulated two simple multi-agent scenarios *EXP1* and *EXP2*. Each agent's dynamics were an instantiation of an SDE of the form of Eq. 1. Collision avoidance during planning was aimed to be achieved with certainty greater than 99%. During collision resolution with the FREE method each agent $\mathbf{a}$ assessed a candidate plan $p^{\mathbf{a}}$ according to cost function $c_{\text{plan}}(p^{\mathbf{a}}) = w_1 \mathbb{E} c_{\text{traj}}(p^{\mathbf{a}}) + w_2 \mathbb{E} c_{\text{miss}}(p^{\mathbf{a}}) + w_3 c_{\text{coll}}(p^{\mathbf{a}})$. Here $c_{\text{traj}}^{\mathbf{a}}$ is a heuristic to penalize the length of the path leading into the goal. Furthermore, $c_{\text{miss}}(p^{\mathbf{a}}) = \max\{0, \mathbb{E}[\|x^{\mathbf{a}}(t_f) - x_f^{\mathbf{a}}\|^2] - (\Delta/2)^2\}$ penalizes deviation from the goal state. The third term $c_{\text{coll}}(p^{\mathbf{a}})$ penalizes collisions (cf. Sec 2.1). The weights are design parameters. For collision resolution we set the weights to $w_1 = .1, w_2 = 100$ and $w_3 = 10^5$, emphasizing avoidance of mission failure and collisions. For bid computation we chose $w_1 = .1, w_2 = 100$ and $w_3 = 0$.

Note, if our method was to be deployed in a receding horizon fashion, the parameters could also be adapted employing standard online learning techniques.

EXP1. Collision resolution was done with the WAIT method to update plans. Draws from the SDEs with the initial plans of the agents are depicted in Tab. 2.2 (left). The curves represent 20 noisy trajectories of agents 1 (red) and 2 (blue). Each curve is a draw from the stochastic differential dynamics obtained by simulating the execution of the given initial plan. The trajectories were simulated with the Euler-Maruyama method for a time interval of $I = [0s, 2s]$. The spread of the families of curves is due to the random disturbances each agent's controller had to compensate during runtime.

Agent 1 desired to control the state from start state $x_0^1 = (5, 10)$ to goal $x_f^1 = (5, 5)$. Agent 2 desired to move from start state $x_0^2 = (5, 0)$ via intermediate goal $x_{f_1}^2 = (5, 7)$ (at time 1s) to final goal state $x_{f_2}^2 = (0, 7)$. While the agents meet their goals under the initial plans, their execution would imply a high probability of colliding around state $(5, 6)$ (cf. Tab. 2.2 (left), Tab. 3). Coordination with fixed priorities (1 (red) $>$ 2 (blue)) yields conflict-free plans (Tab. 2.2 (centre)). However, agent 2 is forced to wait too long at its start location to be able to reach intermediate waypoint $x_{f_1}^2$ in time and therefore, decides to move directly to its second goal. When our definition of social cost penalizes deviation from a goal state highly (via w_2 defined above) this failure to meet the first goal would be reflected in high social cost (cf. Tab. 3). By contrast, the auction method was flexible enough to reverse the

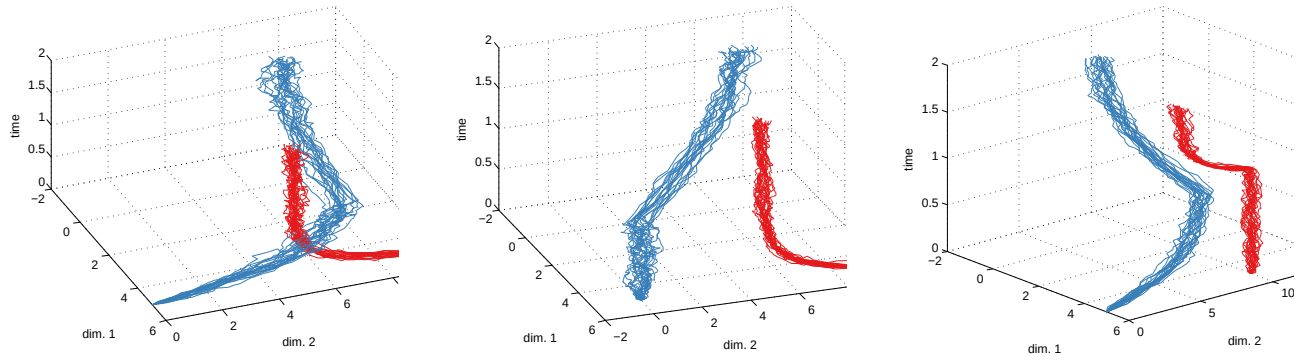


Table 1. EXP1. Draws from uncoordinated Agents’ plans (left), after coordination and collision resolution with methods FP-WAIT (centre) and AUC-WAIT (right).

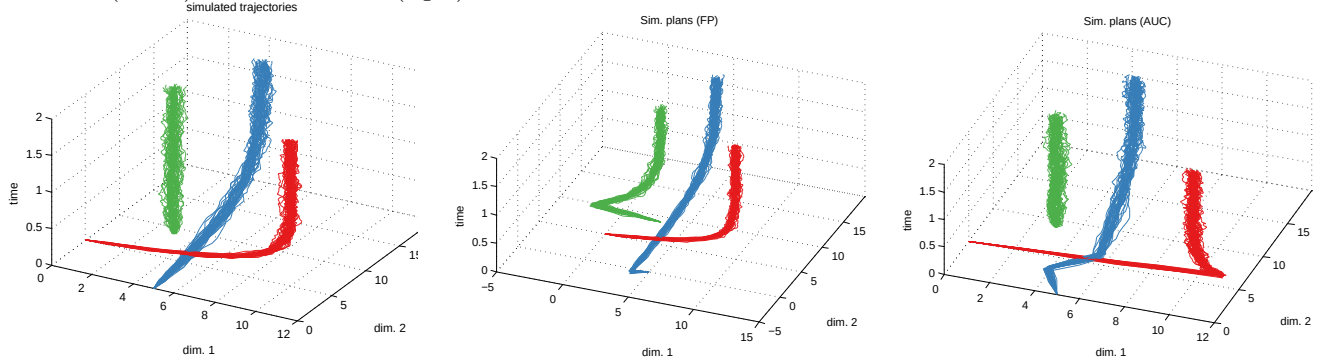


Table 2. EXP2. Draws from uncoordinated Agents’ plans (left), after coordination and collision resolution with methods FP-FREE (centre) and AUC-FREE (right).

ranking at the detected collision point causing agent 1 to wait instead of 2 (Tab. 2.2 (right)). Thereby, agent 2 was able to reach both of its goal states in time.

EXP2. The setup was analogous to EXP1 but with three agents and different start and goal states as depicted in Tab. 2.2. Furthermore, collisions were avoided with the FREE method with three random initializations of the local optimizer. Coordination of plans with fixed priorities (1 (red) > 2 (blue) > 3 (green)) caused 2 to avoid agent 1 by moving to the left. Consequently, 3 now had to temporarily leave its start and goal state to get out of the way (ref. Tab. 2.2 (centre)). With two agents moving to avoid collisions the resulting paths lengths were relatively high (ref. Tab. 3). During coordination with the auction-based method agent 2 first chose to avoid agent 1 (as in the FP method). However, losing the auction to agent 3 at a later stage of coordination, agent 2 decided to avoid three by arcing to the right. This allowed 3 to stay in place yielding lower, overall mean path length (ref. Tab. 3).

Discussion. Our collision detection and resolution

modules succeeded in avoiding collisions. Furthermore, the FREE method for collision resolution always succeeded in generating plans causing the agents to reach their goal locations via collision-free trajectories. Compared to fixed priorities, the flexibility of the auction method seemed to improve the quality of the coordination result but required more overall computation in our experiments. However, parallelization should reduce run-time significantly. Future work will have to investigate benefits and possible shortcomings of the auction-based method more systematically. Additionally, the effects of utilizing a conservative distribution-independent bound as the basis for our collision detection and avoidance modules is visible in our experiments. Using a less conservative bound may cause a few more collisions to occur but could spawn benefits in terms of path length and the number of required coordination rounds.

4. Future Work

Future work will focus on the following questions: How can we quantify the probabilities that all required minimizations succeed given a fixed computational budget? In addition, we will assess our method in systems

Recorded quantity	Experiment 1			Experiment 2		
	NONE	AUC-WAIT	FP-WAIT	NONE	AUC-FREE	FP-FREE
<i>est.Pr[collision][%]</i>	78	0	0	51	0	0
<i>averaged path length away from goal</i>	13.15	13.57	12.57	14.94	16.22	18.13
<i>averaged sqr. dist. of final state to goal</i>	0.05	0.04	25.8	0.05	0.05	0.05
<i>no. of coordination rounds</i>	0	6	3	0	4	4

Table 3. Quantities estimated based on 100 draws from SDEs simulating executions of the different plans in EXP1 and EXP2. Notice, our collision avoidance methods succeed in preventing collisions. In EXP1 the FP-WAIT method failed to reach its first goal in time which is reflected in the *sqr. distance to goal* measure. Note the discrepancies in avg. path length are relatively low due to convexity effects and the contribution of state noise to the path lengths.

with state-dependent noise. We anticipate to be able to speed up our method significantly by means of improved implementation, hardware and parallelization. Finally, we investigate the impact of tighter bounds for collision detection on social cost in settings where such bounds are available.

Regarding our auction mechanism we expect to gain further improvements in terms of social cost by holding auctions between teams of agents rather than individuals. Of course, this raises the concern of coalition formation for improved bidding under computational constraints. Also, we need to investigate whether there are situations where our auction method fails to generate solutions meeting the prescribed goals, does not terminate or is outperformed by the fixed-priority approach. Moreover, in this work, we focus on a cooperative control setting. The bidding protocol merely serves as a computational tool to improve social cost of the overall planning solution in a computationally tractable manner. Are there applications with continuous goods and dynamics as we consider where agents are selfish? How can we modify our bidding rules to yield incentive-compatibility? Or, how can we bring to bear existing strategy-proof mechanisms in such continuous settings?

References

- Ayanian, N. and Kumar, V. Decentralized feedback controllers for multiagent teams in environments with obstacles. *Robotics, IEEE Transactions on*, 26 (5):878–887, 2010.
- Bennewitz, M., Burgard, W., and Thrun, S. Exploiting constraints during prioritized path planning for teams of mobile robots. In *IROS*, 2001.
- Calliess, J., Lyons, D., and Hanebeck, U. Lazy auctions for multi-robot collision avoidance and motion control under uncertainty. Technical Report PARG-01-11, Dept. of Engineering Science, University of Oxford, 2011.
- Erdmann, M. A. and Lozano-Perez, T. On multiple moving objects. *Algorithmica*, 1987.
- Kostic, D., Adinandra, S., Caarls, J., and Nijmeijer, H. Collision-free motion coordination of unicycle multi-agent systems. In *American Control Conference (ACC), 2010*, pp. 3186–3191. IEEE, 2010.
- Lagoudakis, M., Markakis, V., Kempe, D., Keskinoak, P., Koenig, S., Kleywegt, A., Tovey, C., Meyerson, A., and Jain, S. Auction-based multi-robot routing. In *Int. Conf. on Robotics: Science and Systems*, 2005.
- Lyons, D., Calliess, J., and Hanebeck, U. Chance constrained model predictive control for multi-agent systems with coupling constraints. In *American Control Conference (ACC)*, 2012.
- Tovey, C., Lagoudakis, M., Jain, S., and Koenig, S. The generation of bidding rules for auction-based robot coordination. In *Multi-Robot Systems: From Swarms to Intelligent Automata*, volume 3, pp. 3–14, 2005.
- Whittle, P. A multivariate generalization of Tchebichev’s inequality. *Quarterly Journal of Mathem.*, 1958.